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DevOps History

By bringing software development and IT operations together, DevOps enables teams to deliver software changes more often and more reliably. GeeksforGeeks says DevOps is well-suited to the software lifecycle because it combines collaboration, automation, and continuous software development. DevOps didn't emerge at the same time, but its history is closely tied to ideas such as the Lean Movement, the Agile Manifesto, and continuous Delivery. Each added to the contribution by improving how software works and moving from an idea to something users can use.

The reason the Lean movement became so important is its origins in manufacturing practices, especially Toyota's production system. Toyota uses a system that stops the line when a problem is detected, leveraging just-in-time production and jidoka to prevent defects from being passed along by accident. Lean focuses on reducing waste and delivering the highest quality to customers. Lean focuses on what customers want and how to move through a process, using principles that encourage teams to reduce delays, respond to high demand, and continually improve their work as the field grows (Do, 2017). A value stream includes the steps you need to deliver a product or service, which range from requesting a feature to releasing it. A feature might spend more time waiting for review, testing, or approval than a team spends developing it, so looking at the whole process can help these teams identify delays and unnecessary handoffs more quickly. Lean influenced DevOps by encouraging teams to improve the entire delivery

process, because if each department focused only on its own work, delays could build up departments and put them behind. The idea of Lean encourages people to see how their work affects the next step and the customer. It also focuses on small improvements that align with DevOps practices, such as reviewing and refining processes over time (Do, 2017; Humble).

The Agile Manifesto was put together by a team of seventeen software practitioners in 2001, who met to discuss ways they could improve software development. Individuals and interactions over process over tools, working software over comprehensive documentation, customer collaboration over contract negotiations, and responding to change over following a plan are four values the team came up with (Agile Alliance). These values encourage development teams to deliver working software in small increments and to use feedback to guide their future work. Agile helps teams respond to user feedback sooner, rather than waiting until the end of a long project to show results. Communication between developers and business representatives is important, and the values encourage it (Agile Alliance).

Agile introduced improvements in how teams develop software, even though it did not always address what happens after development, because a completed feature could stall while waiting for testing, operations work, or a scheduled release (Humble).

Continuous Delivery helps teams keep software ready to release by making small changes and checking them consistently. Teams use automation to handle repetitive work so they can focus on fixing problems and improving the software. Everyone shares responsibility for ensuring the project works correctly (Humble). A good example is when automated tests can check a small code change for problems; if something goes wrong, the team can easily find the cause because there are fewer changes to look through (Humble). Continuous Delivery doesn't mean every change is automatically available to users; instead, the software stays ready until the

team decides to release it. Lean aligns well with this concept by reducing those delays, and Agile helps teams get feedback more often (Agile Alliance; Humble).

The DevOpsDays website says 2009 was the year of its first event in Ghent, Belgium and then founder, Patrick Debois, had given a presentation on his ideas about Agile infrastructure the year before in Toronto (DevOpsDays). They create a place for people working in development and operations to discuss how they could work together (DevOpsDays). These ideas came together around the same delivery problem that developing software is only one part of getting value to customers. Teams should also ensure the software works well before release and keep it running smoothly (GeeksforGeeks, 2026).

DevOps carries Lean's focus on improving the flow of work, using Agile's feedback and Continuous Delivery's automation, into the relationship between development and operations. When teams share responsibility for the entire process, they can catch problems earlier, and users can depend on their releases (Do, 2017; Humble).

The history of DevOps connects to several movements that help improve how work gets done: Lean, which offers ways to reduce waste and improve flow, and the Agile Manifesto, which encourages teams to deliver working software. They also have continuous delivery automation to apply changes, and small batches are used to prepare software for release. DevOps brought these ideas together across development and operations to create a safe environment for users and to understand the connections that explain why DevOps involves both tools and teamwork (Agile Alliance; Do, 2017; Humble).

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